

YEKATERINO-NIKOL'SKOYE, Russian Federation

WMO: 317070

Lat: 47.7447N

Lon: 130.9703E

Elev: 72

StdP: 100.46

Time Zone: 10.00 (E10)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -29.1	(c) -27.0	(d) -32.1	(e) 0.2	(f) -28.4	(g) -30.3	(h) 0.2	(i) -26.1	(j) 16.0	(k) -15.9	(l) 12.3	(m) -19.2	(n) 0.9	(o) 320	(p) 0.506

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 8.6	(c) 30.1	(d) 22.2	(e) 28.7	(f) 21.6	(g) 27.3	(h) 20.9	(i) 24.4	(j) 27.7	(k) 23.2	(l) 26.5	(m) 22.2	(n) 25.5	(o) 1.8	(p) 200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
23.4	18.4	26.5	22.2	17.1	25.1	21.1	15.9	24.2	74.5	27.8	69.9	26.6	66.0	25.5	29.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10.1	7.7	6.3	-32.2	33.1	2.8	1.8	-34.2	34.4	-35.9	35.5	-37.5	36.5	-39.5	37.8
			-31.8	25.7	2.8	1.6	-33.8	26.9	-35.5	27.8	-37.0	28.8	-39.0	29.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	3.1	-18.6	-14.2	-5.4	5.3	13.3	18.6	22.0	20.3	14.1	4.9	-7.2	-17.2
	DBStd	15.01	4.11	4.53	5.72	4.11	3.90	3.39	2.33	2.67	3.54	4.28	6.22	4.37
	HDD10.0	3736	886	678	477	151	13	0	0	0	7	164	516	844
	HDD18.3	5812	1144	911	736	392	163	37	3	11	133	416	766	1102
	CDD10.0	1211	0	0	0	9	114	259	372	320	129	7	0	0
	CDD18.3	246	0	0	0	0	6	46	117	72	5	0	0	0
	CDH23.3	1772	0	0	0	7	121	397	777	425	44	1	0	0
	CDH26.7	431	0	0	0	1	24	123	204	74	5	0	0	0
Wind	WSAvg	1.9	2.4	2.0	2.2	2.3	2.2	1.8	1.5	1.4	1.5	1.8	2.0	2.2
Precipitation	PrecAvg	625	4	5	10	24	57	98	136	156	79	31	17	10
	PrecMax	898	13	22	30	162	140	241	313	332	206	119	62	28
	PrecMin	329	0	0	0	0	3	9	21	19	10	1	0	0
	PrecStd	156	4	6	6	27	31	51	65	82	53	26	16	7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-6.3	2.0	12.1	24.1	29.3	33.1	32.0	30.2	26.9	20.7	10.0	-3.5
		MCWB	-7.8	-1.5	5.7	14.0	17.5	21.3	23.5	24.2	20.1	12.2	5.6	-5.1
	2%	DB	-8.9	-1.6	8.5	19.2	26.6	29.8	30.3	28.6	24.3	17.2	5.9	-6.7
		MCWB	-10.2	-4.2	3.6	10.6	15.7	20.8	22.8	22.8	17.9	10.4	2.5	-8.0
	5%	DB	-10.8	-4.4	5.7	16.1	24.1	27.5	29.0	27.2	22.6	14.6	3.1	-8.9
		MCWB	-11.9	-6.2	1.5	8.6	14.7	19.7	22.4	21.9	16.9	8.8	0.3	-10.0
	10%	DB	-12.5	-6.8	3.1	13.4	21.6	25.3	27.6	25.8	20.7	12.0	0.9	-10.7
		MCWB	-13.4	-8.4	-0.2	7.0	13.5	18.8	21.9	21.0	15.8	7.2	-1.0	-11.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-7.4	-1.4	6.2	14.9	19.3	23.7	26.2	25.8	21.4	13.8	6.3	-4.7
		MCDB	-6.2	1.6	10.7	23.4	26.3	29.6	29.6	29.0	24.5	19.2	9.8	-3.7
	2%	WB	-10.2	-4.0	3.9	11.2	17.4	22.1	24.7	24.3	19.4	11.4	2.7	-7.9
		MCDB	-9.1	-1.9	8.2	18.0	23.7	27.2	28.0	26.8	22.3	15.8	5.4	-6.9
	5%	WB	-11.8	-6.1	1.6	9.3	15.8	20.8	23.6	23.1	18.1	9.3	0.7	-9.9
		MCDB	-10.8	-4.3	5.0	15.2	22.1	25.8	27.1	26.0	21.1	13.5	2.6	-9.1
	10%	WB	-13.4	-8.3	0.1	7.6	14.3	19.6	22.7	21.9	16.7	7.5	-0.7	-11.6
		MCDB	-12.5	-6.9	3.0	12.3	20.4	24.1	25.8	24.5	19.7	11.4	1.1	-10.7
Mean Daily Temperature Range	5% DB	MDBR	9.5	11.4	11.1	11.5	11.5	9.5	8.6	9.0	11.0	10.7	8.7	8.6
		MCDBR	10.7	12.8	13.0	17.2	16.6	14.1	11.5	11.2	13.8	15.2	11.2	9.5
		MCWBR	9.5	10.7	8.6	9.9	7.7	6.1	5.3	5.7	7.4	8.5	7.9	8.4
	5% WB	MCDBR	10.7	12.7	12.3	15.5	14.5	12.0	9.5	9.2	11.2	14.3	10.4	9.2
		MCWBR	9.6	10.6	8.3	9.5	7.6	6.1	5.6	5.7	6.9	8.8	7.5	8.2
Clear-Sky Solar Irradiance	taub	0.296	0.330	0.414	0.540	0.581	0.542	0.516	0.465	0.426	0.435	0.357	0.305	
	taud	2.145	2.041	1.876	1.730	1.716	1.905	2.022	2.153	2.161	1.985	2.094	2.117	
	Ebn at Noon	753	802	778	714	705	740	755	778	769	667	656	680	
	Edh at Noon	85	119	167	214	225	188	165	139	125	127	91	78	
All-Sky Solar Radiation	RadAvg	1.95	3.20	4.49	4.78	5.27	5.50	5.19	4.79	4.00	2.73	1.81	1.50	
	RadStd	0.13	0.10	0.35	0.50	0.48	0.72	0.55	0.32	0.39	0.20	0.13	0.10	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (6 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page